MATERIALS DATA · #9 GRAPHITE

You think graphite is pencils. It is the biggest thing in your battery.

Graphite is the anode in every lithium-ion battery, the largest mineral by mass in an EV pack. Pencils use barely a scrap of it. Here is the real story in data, and why it runs through one country.

Swipe, a materials engineer's data tour ⟶

100 kg

Of graphite in one EV battery, more than every other battery metal combined.

Nickel, cobalt, manganese and lithium together come to about **59 kg**. Graphite alone is around **100 kg**, roughly 250,000 pencils' worth. It is the heaviest ingredient in the battery and the one people notice least.

Not pencils. Power.

28%

of graphite demand now goes to battery anodes, the fastest-growing use, up from almost nothing a decade ago.

62%

is where batteries are headed by 2036 on current forecasts. Graphite is becoming a battery material first.

Pencils and lubricants are a rounding error. The story of graphite this decade is the anode.

One mined, one baked.

Natural graphite mined as flake or amorphous rock	**~35%**
Synthetic graphite baked from petroleum coke and coal tar pitch	**~65%**

About two-thirds of the graphite the world uses is made, not mined: cooked from oil-refining leftovers at very high heat. Both end up as the same carbon anode.

Batteries now, steel still.

Battery anodes lithium-ion cells	**~28%**
Refractories and steelmaking crucibles and furnace linings	**~26%**
Graphite electrodes electric-arc furnace steel	**~16%**

For a century graphite was a steel material: crucibles, linings, electrodes. Batteries have only just become its largest single use.

One country, the whole chain.

Mine production natural graphite, 2024	**78%**
Refining and processing share of world capacity	**90%**
Anode material battery anode-grade graphite	**93%**
Spherical graphite battery-ready spheronised	**99%**

China's grip tightens at every step. The further downstream you go, the closer it gets to all of it.

The chokehold is the processing.

You can mine graphite in Madagascar, Brazil or Mozambique. Turning flake into the rounded, 99.95% pure **spherical graphite** a battery needs is the hard, dirty, capital-heavy step, and China does almost all of it.

In December 2023 China put graphite under **export licensing**. Flake exports fell about 25% the next year. The leverage is real.

The anode you cannot skip.

Every mainstream lithium-ion battery needs a graphite anode, and there is no drop-in replacement at scale today. Silicon helps at the margin, but graphite still carries the cell.

So the quiet pencil material became a battery chokepoint. For anyone building EVs or grid storage outside China, that single fact is the whole problem.

Sources: USGS Mineral Commodity Summaries 2025 (graphite); Benchmark Mineral Intelligence and IDTechEx for the anode supply chain.

Next up: silicon, the anode of tomorrow and the chip of today.

I'm a materials engineer turned data analyst, breaking down one material at a time: the numbers, the science, and why it matters. Steel, aluminium, copper, rare earths, lithium, nickel, cobalt and titanium came first. Save this, and follow for silicon.

Interactive dashboard + data on GitHub

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